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Stopped-light-enhanced gravitational force sensing

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As the weakest fundamental force, gravity yields only small accelerations from millimetre-scale masses, making them difficult to detect. Detecting the gravitational fields of microscopic objects is important for microscale sensing, imaging or studies of the impact of the quantum state of an object on gravity. Cavendish-type torsion balances can sensitively measure gravity from small source masses, but their integration with micro- and nanophotonic readout remains challenging. Here we show gravitational-force sensing of a millimetre-sized mass using a torsion pendulum integrated with optical microcavity readout. A coupled photon–phonon resonance induces stopped light behaviour, reducing the group velocity to 2,000 m s^{-1} and producing steep optical dispersion. The exponentially decaying evanescent field of the microcavity converts nanoscale displacements over an ~100 nm coupling length into optical signals, thereby increasing displacement sensitivity. The sensor reaches a displacement sensitivity of 7.85 pm/$\sqrt{\text{Hz}}$, corresponding to an acceleration sensitivity of 3.06×10^{-16} g. By periodically modulating the mass position, we resolve a minimum detectable gravitational-force change of 3.02×10^{-16} N. This compact, sensitive scheme supports microscopic gravitational sensing and may allow for studies of gravity's quantum nature.

Gravity is weak and remains outside both the Standard Model and an established quantum description. Since the pioneer work[1], tests have ranged from gravitational-wave observations and black-hole imaging[2,3], together with strong-field calculations[4], to laboratory measurements of Newton's constant[5–7] and Newtonian gravity at micrometre separations[8–10]. These laboratory studies generally use kilogram-scale source masses[11–13]. Measuring the field of a quantum-prepared source in a 'quantum Cavendish experiment'[14–19] would provide a tabletop probe of gravity in the quantum regime. However, measuring gravitational forces from increasingly smaller source masses is challenging, as the force generated by a spherical mass diminishes rapidly with the cube of its radius. This difficulty has limited experimental observations involving small masses[20].

Weak gravitational forces are difficult to detect, and optical, mechanical and electrical resonances have been used to enhance weak-force measurements[21–27]. Cavity optomechanics reads mechanical motion through changes in an optical resonance and can operate near the standard quantum limit. Such resonance-enhanced detections manifest themselves in various critical fields, such as quantum gravimeters, accelerometers, weak-force sensing and the Casimir effect of vacuum fluctuation[28–32]. Detecting weak gravitational forces, especially from small masses, is fundamentally limited by thermally induced amplitude fluctuations (Brownian force noise) in sensors[33,34]. A torsion-balance experiment[35] periodically moved the source mass and detected the resulting time-dependent Newtonian signal. Similarly, coherent photon–phonon interference has enabled

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optomechanically induced transparency, absorption and slow/fast light in whispering-gallery-mode (WGM) resonators, nanoscale optomechanical crystals and membrane-in-the-middle cavities[36–38]. Related electromechanical platforms based on mass-loaded membranes coupled to microwave cavities have also been proposed and developed for future gravitational-force measurements in a possible quantum regime[39,40]. These studies provide the optomechanical and electromechanical context for the Brillouin-scattering-induced absorption (BSIA)-induced stopped-light sensing investigated here. Small-scale gravitational sensing has been advanced through miniature torsion balances, cryogenic levitated mechanical sensors and mesoscopic particle gravimetry protocols[20,27,35]. In parallel, WGM resonators have provided highly sensitive optical transduction of mechanical displacement and acceleration[41,42]. Further enhancement of its much-anticipated sensing ability may be achieved with integrated micro- or even nanomechanics. However, it is a challenge to construct a detection scheme or apparatus scale from millimetre-sized objects down to nanometre-scale sensing units until the current work.

In this paper, we introduce a gravity and acceleration sensing platform based on exploiting the steep optical dispersion created at a stopped-light point (SP) within a chip-scale, 150-μm-radius optical microcavity. The sensing mechanism leverages a photon–phonon hybrid resonance—specifically, BSIA to generate a steep phase gradient that enhances optomechanical interactions, thereby boosting sensitivity to mechanical displacement. We experimentally realize this enhancement in sensing gravitational force and acceleration of a millimetre-sized mass based on a hierarchical torsion pendulum. The device achieves the minimum displacement of 7.85 pm/$\sqrt{\text{Hz}}$, resolving a minimum detectable acceleration of 3.06×10^{-16} g and a minimum gravitational-force change of 3.02×10^{-16} N from a millimetre-sized mass. This compact, room-temperature scheme supports high-precision gravimetry and applications in fundamental physics and quantum sensing.

Torsion-pendulum sensing architecture

The torsion pendulum transfers gravitational coupling between the source and test masses to a silica microsphere microcavity (~150 μm radius) and tapered-fibre optical readout that resolves sub-nanometre displacement (Fig. 1a). An ~100 nm microsphere–taper gap evanescently couples the signal laser into the microcavity. A 1.2 g, 6-mm-diameter source mass (m_s) is placed ~4.2 cm from the test mass (m_t) behind an aluminium-foil Faraday shield that suppresses electrostatic force. The Faraday shield was tested with an external 100 Hz electric field source at the source-mass side. With the shield grounded, the interference was attenuated by 35 dB (spectrum amplitude from −93 dBm to −128 dBm), confirming effective suppression of external electric coupling. The shield did not affect optical transmission or the vector network analyser (VNA) noise floor (details shown in Supplementary Section 13). A piezoelectric stage reciprocates the source mass, producing a time-modulated field that deflects the torsion pendulum and changes the microcavity–taper distance d. Owing to the exponential decay of the evanescent field within the coupling range (~100 nm), any small change in the coupling distance can alter the coupling rate, and thus the output light intensity can be used to perceive changes in acceleration. The apparatus was operated with the acrylic enclosure closed and the optical-table air suspension activated. In an impulsive floor-disturbance test, the air suspension reduced the interferometric vibration amplitude from 0.0462 V to 0.00477 V, corresponding to 19.7 dB attenuation. The reported noise-equivalent acceleration was measured under this normal operating condition and therefore includes the residual environmental vibration transmitted to the sensor. Apparatus photographs are shown in Extended Data Fig. 1, and environmental-control measurements are provided in Supplementary Section 14.

Stopped-light dispersion enhancement

This gravitational-force sensing can be enhanced by the steep dispersion, that is, stopped light[43], in an optical microcavity. A strong control field at ω_C enters through the tapered fibre and drives forward stimulated Brillouin scattering in the microcavity (Fig. 1b), creating a Stokes optical WGM together with an acoustic WGM at Ω_B (ref. 44). When the probe ω_P is tuned near the Stokes resonance ω_{SO}, it couples coherently to the acoustic mode. This interaction produces BSIA, analogous to electromagnetically induced absorption in atomic and optical-resonator systems[45–49] (Fig. 1c). Specifically, BSIA gives rise to a group delay of about 420 μs, reducing the light's group velocity to approximately 2,000 m s^{-1} while enhancing the system's optical dispersion. This narrow BSIA window induces a steep transmission phase response (Fig. 1d)—the defining feature of the SP, where phase–frequency slope diverges to maximize perturbation-to-signal conversion. External gravitational perturbation drives torsional deflection, shifting the optomechanical mode; the SP's steep phase dispersion amplifies this tiny shift into a measurable optical signal—validating the BSIA-induced SP as a photonic platform for weak perturbation sensing. Enhancement combines the SP's steep phase–frequency slope with evanescent microcavity–taper conversion of nanogap displacement into coupling-rate modulation.

Unlike traditional dispersion control methods, our scheme features a dispersion singularity—'stopped-light point' [43]—defined by a divergent phase–frequency slope that maximizes the system's responsiveness to external perturbations. This singularity occurs at the transition between slow-light (normal dispersion) and fast-light states (anomalous dispersion). The key attribute, in addition to the zero group velocity itself, is the phase sensitivity caused by the divergent phase slope. Such an SP exhibits maximized group delay, enabling dispersion-enhanced sensing. Here we experimentally selected a Stokes optical mode with a redshift of 200.9 MHz and observed an absorption peak under the Brillouin phase-matching condition (Fig. 2a). The measured acoustic linewidth of 12.3 kHz indicates that the system has strong dispersion capability. On the basis of the previous work[43], the group delay ($\tau \sim d\phi/d\omega$) can be calculated by the formula (1):

$$\begin{aligned}\tau^{(T)}|_{\omega_P=\omega_C-\Omega_B} &= -\frac{d\phi}{d\omega} = R\left\{\frac{-i}{T(\omega_P)}\frac{dT(\omega_P)}{d\omega_P}\right\} \\ &= \kappa_S\left(1+\frac{g^2}{\Gamma_B^2}\right)\cdot\frac{1}{C^2-\kappa_S C-\delta^2}\end{aligned} \quad (1)$$

The denominator contains the quadratic term $C^2 - \kappa_s C - \delta^2$; its derivation is provided in Supplementary Section 1. Here $C = (g^2 - Gcy_B\kappa_s)/Gcy_B$ denotes the effective gain cooperativity, while κ_s and Gcy_B characterize dissipation in the Stokes optical resonance and the acoustic resonance. The BSIA detuning is $\delta = (\omega_c - \Omega_B) - \omega_{SO}$, and the nonlinear coupling $g = |\beta A_c|$ is determined by the stimulated Brillouin scattering coefficient β and intracavity control field A_c. When the denominator vanishes, τ diverges and $v_g \sim 1/\tau$ approaches zero, which defines the SP. Figure 2c–e presents the BSIA transmission, phase and group delay for five detunings. At the initial detuning, $\delta^2 > C^2 - \kappa_s C$, the negative group delay places the system in the fast-light regime. Reducing δ deepens and narrows the BSIA feature while increasing the magnitude of the phase slope and group delay. At $\delta^2 = C^2 - \kappa_s C$, the denominator becomes zero, giving the SP with a divergent group delay and maximum phase slope. With a further reduction in δ, the phase slope changes sign and the group delay becomes positive ($\delta^2 < C^2 - \kappa_s C$), before decreasing within the slow-light regime. As δ is scanned, the denominator in equation (1) therefore changes sign twice; the SP is reached when $\tau = \infty$. Figure 2b plots the resulting transition between the fast- and slow-light regimes.

Acceleration sensing performance

To verify the BSIA-induced SP's photonics-enabled sensing enhancement, we first characterized acceleration sensing performance.

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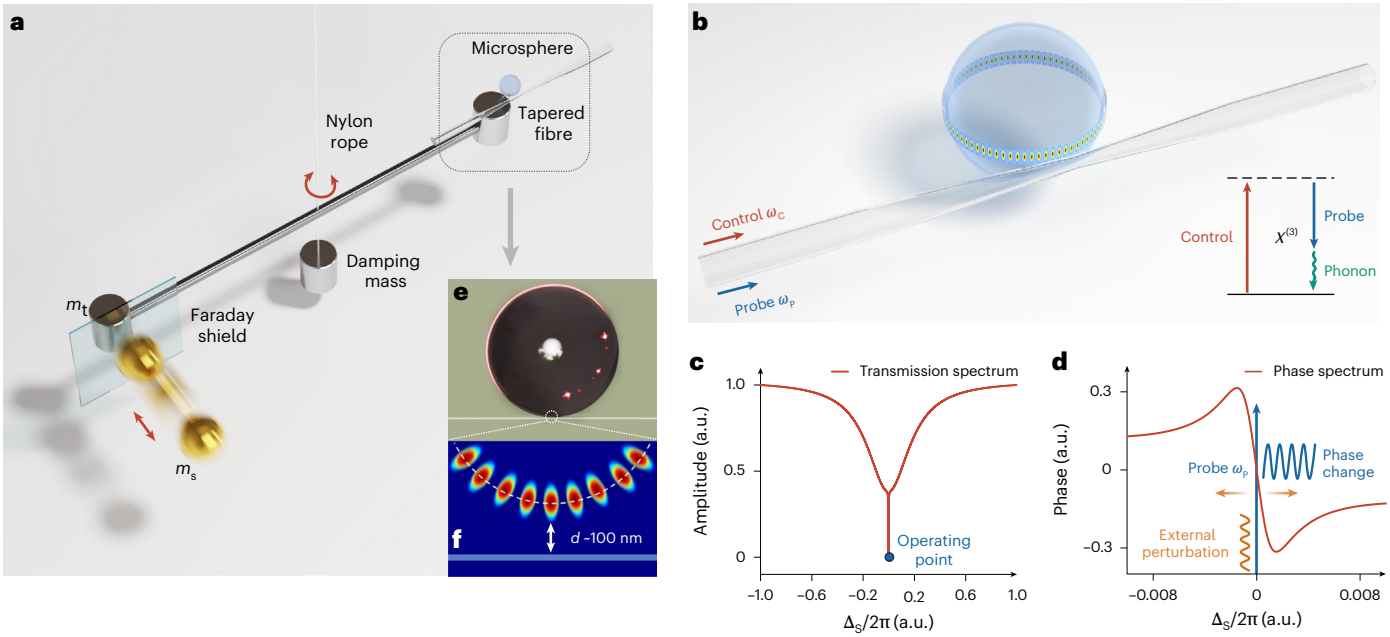


Fig. 1 | SP-enhanced gravitational acceleration sensing. a, Experimental schematic of the torsion-pendulum-based gravity acceleration sensor. The pendulum consists of two stainless-steel columns separated by 400 mm and connected by a rigid optical rod. A nylon suspension fibre provides a well-defined torsional mode, isolated from other mechanical degrees of freedom. One column carries a test mass (m_t = 100 g), while the opposite column supports a silica microsphere optical cavity (mass ≈ 31.1 µg, radius r ≈ 150 µm), which serves as the sensing element (dashed box) and is elastically supported by a lightweight optical fibre. A damping mass (100 g) is suspended below the rod centre to enhance stability and accelerate mechanical recovery. A lead bead gravitational source (m_s = 1.2 g, r ≈ 3 mm), mounted on a piezoelectric translation stage, provides a controllable time-modulated gravitational force. The resulting torsional deflection displaces the microsphere relative to a tapered fibre, modulating the evanescent coupling and encoding the gravitational signal. A 50-µm-thick Faraday shield suppresses electrostatic interactions. Pendulum deflections vary the coupling distance d by ~100 nm, producing an exponential change in the external coupling rate. **b**, Principle of Brillouin scattering. A control field (ω_C) excites a forward Brillouin acoustic mode (Ω_B) via the $\chi(3)$ nonlinearity under phase-matching conditions ($\omega_P = \omega_C - \Omega_B$). A probe field ($\omega_P$) interrogates the Stokes optical mode. **c**,**d**, Transmission (**c**) and phase (**d**) spectra of BSIA. When the probe is tuned to the minimum of the BSIA dip (marked operating point), the phase exhibits maximal dispersion. External forces induce BSIA frequency shifts through the optomechanical coupling identified in **d**, modifying the beat phase between ω_P and ω_C. At the SP, where the phase slope is maximal, the interferometric phase response is most sensitive to external perturbations. Insets **e** and **f** show an optical micrograph of the microsphere–taper system and the optical-field distribution in the coupling region, respectively; the latter indicates the ~100 nm microsphere–taper gap.

The experimental set-up (inset of Fig. 3a) consists of a microcavity attached to a stainless-steel column, mounted on a piezoelectric displacement stage to generate controlled acceleration—directly probing the SP's phase dispersion effect on optomechanical coupling. The stage was controlled to drive the microcavity in a reciprocal motion perpendicular to the tapered fibre. Through passive locking and balance detection scheme[47] under various detunings near the SP, we tested the temporal response (contour map) to a constant acceleration of $a = 0.6 \times 10^{-12}\,g$ (g = 9.8 m s^{-2}, at 0.625 Hz) as shown in Fig. 3b. An increase in the signal amplitude has been observed as δ increased, reaching a maximum peak-to-peak value at SP, which corresponds to the SP condition (middle area). For five δ conditions, Fig. 3a obtains sensitivity from the slope of output voltage versus acceleration, varied through the piezoelectric-stage voltage. The maximum sensitivity (~10^8 V g^{-1}) occurs at the SP, validating enhancement by its divergent phase slope. The higher peak-to-peak output voltage at the SP (Fig. 3b) further corroborates that steep phase dispersion, as the core photonics mechanism, is responsible for the sensing enhancement. To distinguish maximum transduction gain from maximum signal-to-noise performance, driven and undriven traces were measured at five detunings under the same $0.6 \times 10^{-12}\,g$, 0.625 Hz excitation and identical detection bandwidth. The peak-to-peak signal-to-root-mean-square-noise ratio reaches 26.7 dB at the SP, 4.7 dB above the nearest off-SP point. Under the present noise spectrum and readout bandwidth, the SP therefore gives the maximum transduction gain and measured signal-to-noise metric among the tested points; this conclusion is specific to this implementation (experimental details and analysis shown in Supplementary Section 11).

To identify the optimum accelerometer frequency, the configuration shown in Fig. 3a was used to record the Brillouin output while the piezoelectric-base frequency was swept from 1 Hz to 10 Hz (Fig. 3c). A synchronous interferometer measurement converted structural displacement to acceleration (Fig. 3d), where the interferometer signal decreased below 4.4 Hz while the Brillouin beat-frequency amplitude remained relatively stable, indicating superior low-frequency performance. The structural power spectral density (PSD) was obtained by Fourier-transforming and power-normalizing the undriven laser-output time series (results in Supplementary Section 4). Frequency-dependent sensitivity was obtained by dividing the Brillouin response by the interferometrically measured acceleration.

After demodulating the interferometer signal, the ratio of the Brillouin output in Fig. 3c to the acceleration in Fig. 3d gives the frequency-dependent sensitivity in Fig. 3e. Figure 3e reaches 10^8 V g^{-1} at low frequencies through mechanical resonance and SP phase dispersion. In the sub-1 Hz range (for manual signal collection, see illustration and Supplementary Section 4b), the system achieves a minimum detectable acceleration of $3.06 \times 10^{-16}\,g$—validating the SP's ability to support low-frequency sensing. Compared with the high-bandwidth accelerometer of ref. 50, which has a milligram-range detection limit, our lower-bandwidth scheme reaches a lower low-frequency limit. Ambient operation without vacuum or cryogenic cooling produces strong near-DC 1/f (pink) noise. Despite this, the SP's steep phase dispersion maintains high sensitivity—demonstrating the robustness of the photonics mechanism under ambient conditions. The VNA

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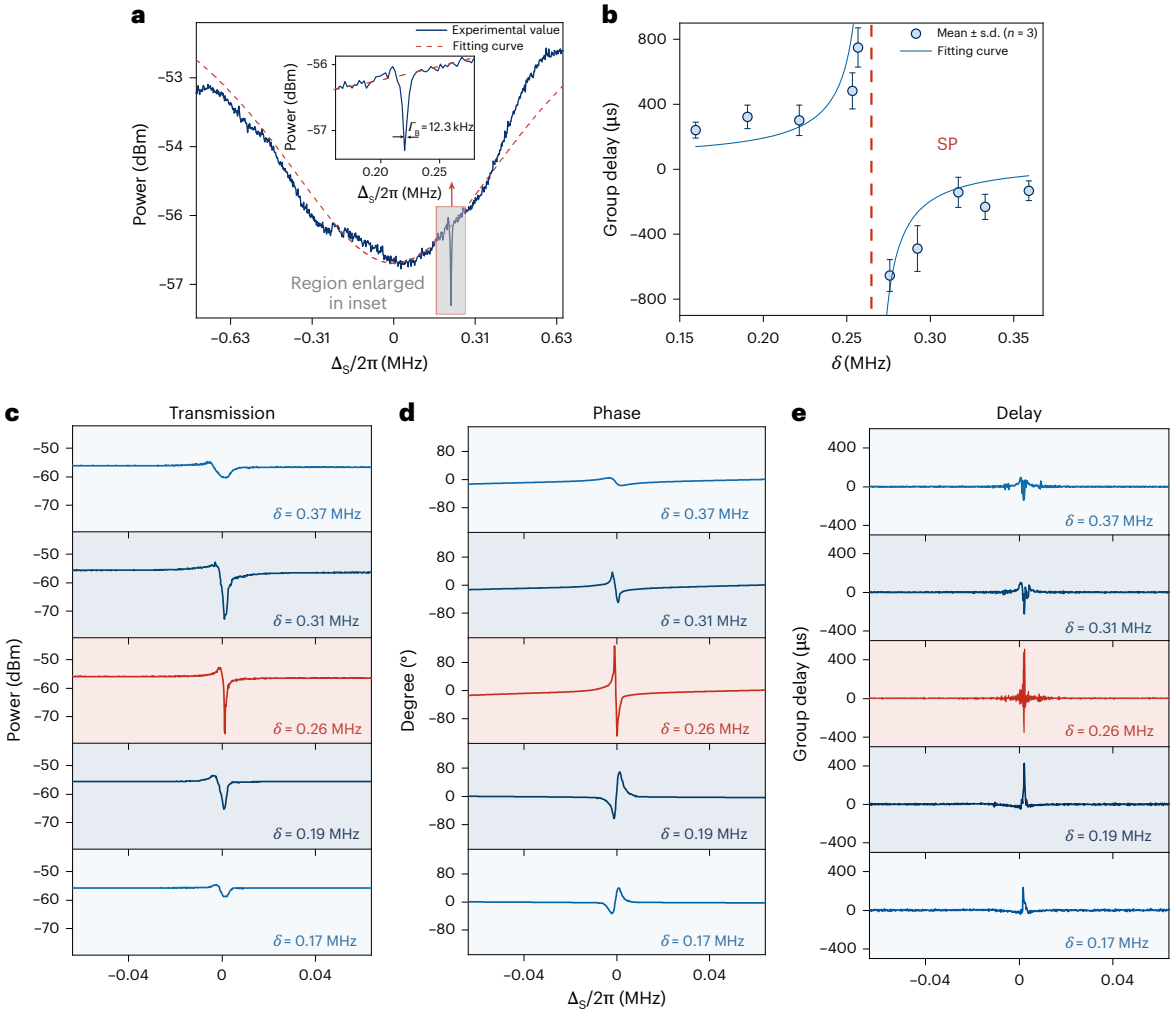


Fig. 2 | Observation of the SP. a, BSIA observed in a Stokes optical mode with a downshifted frequency of 200.92 MHz (centre frequency set to 0 MHz). The optical resonance linewidth is 1.03 MHz, while the acoustic absorption window has a linewidth of 12.3 kHz. The frequency detuning is $\delta = 0.22$ MHz. The shaded frequency interval marks the BSIA feature enlarged in the inset. For **b**, data are presented as mean values ± s.d. from $n = 3$ technical replicate VNA scans at each detuning, all acquired using the same microsphere–taper device; one complete scan constitutes one technical replicate. Technical replicates were used to quantify scan-to-scan repeatability, and no control group is applicable because **b** shows a continuous detuning sweep rather than a comparison between experimental groups. By tuning the control-laser frequency, δ is varied indirectly, driving the system from negative group delay (fast-light regime) to a divergence at $\delta = 0.26$ MHz, where the group velocity approaches zero. Beyond this point, the group delay changes sign and becomes positive (slow-light regime). The divergence at $\delta = 0.26$ MHz marks the SP. **c**–**e**, Transmission spectra (**c**), phase responses (**d**) and group delays (**e**) measured for five representative detunings: $\delta = 0.37$ MHz, 0.31 MHz, 0.26 MHz, 0.19 MHz and 0.17 MHz. As δ decreases towards the SP, the BSIA window deepens and narrows, accompanied by a rapid enhancement of phase dispersion. The corresponding phase slopes are -7.4×10^3 deg Hz^{-1}, -3.8×10^4 deg Hz^{-1}, -3.1×10^5 deg Hz^{-1}, 6.9×10^4 deg Hz^{-1} and 1.1×10^4 deg Hz^{-1}. At the SP ($\delta = 0.26$ MHz), the absorption window is maximally confined, the phase–frequency slope is largest, and the group delay diverges. Crossing the SP reverses the sign of the group delay, identifying the SP as the fast–slow light transition point and the regime of strongest dispersion.

phase-locked loop shifts the mixed signal to an intermediate frequency with a 1 Hz bandwidth, reducing the noise floor. In Fig. 3c,d, dividing the noise-floor PSD by sensitivity gives a noise-equivalent displacement of 7.85 pm/$\sqrt{\text{Hz}}$ (derivations are provided in Supplementary Section 5c).

Gravitational-force measurements

As a demonstration of the SP-based photonic platform's sensing capability, we performed static gravitational-force measurements (analogous to the Cavendish experiment) using the set-up in Fig. 4a, identical to time-modulated measurements except for a fixed source mass, thereby extending the platform from dynamic acceleration to weak static-force sensing. A 1.5 kg lead source was initially 3 cm from the test mass. Torsion moved the microcavity away from the tapered fibre, decreasing the coupling rate and signal; after approximately 24 min, the source–test distance changed by 22 nm, which was captured by the change in the beat frequency interference signal shown in Fig. 4b. Because the evanescent coupling response is nonlinear, the 22 nm displacement at the microcavity readout end was inferred using the full experimentally calibrated exponential relation between the VNA beat-note amplitude and the microcavity–taper gap, rather than the first-order approximation. The measured decay constant was $\alpha = 0.00833\ \text{nm}^{-1}$, for which the first-order approximation would introduce a 1.89% error at 22 nm. The full calibration agreed with the independent interferometric measurement within 5% over the tested displacement range. Slow operating-point variations were monitored through the baseline coupling signal and included in the calibration uncertainty (experimental details and error analysis shown in Supplementary Section 10). This result demonstrates the high resolution of our static measurement approach. To examine our detection sensitivity, we modulated the gravitational field over time. The source mass in Fig. 1 reciprocates towards the test mass from a 4.2 cm centre-to-centre

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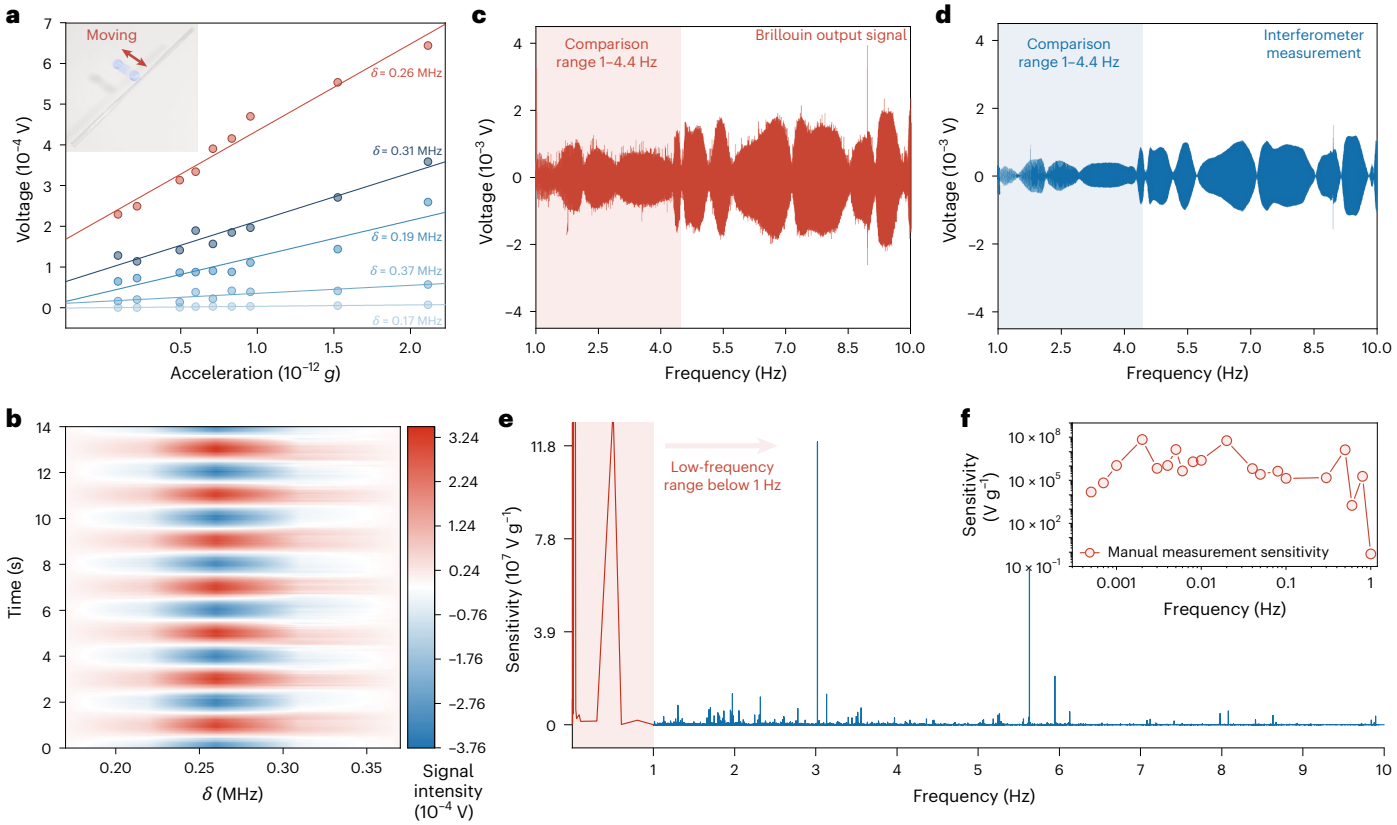


Fig. 3 | SP-enhanced acceleration sensing. a, Output signal voltage as a function of acceleration measured at five different optical detunings (δ), corresponding to Fig. 2c–e. The optical microsphere resonator, mounted on a steel column, is driven by a piezoelectric stage to oscillate perpendicular to the tapered fibre (schematic shown upper left). The extracted sensitivities (slopes) are 3.4×10^6 V g^{-1}, 8.8×10^7 V g^{-1}, 2.1×10^8 V g^{-1} (SP), 1.2×10^8 V g^{-1} and 1.9×10^7 V g^{-1}, with the maximum sensitivity achieved at the SP. **b**, Brillouin output signal contour maps under identical acceleration ($a = 0.6 \times 10^{-12}$ g, $g = 9.8$ m s^{-2}) at 0.625 Hz for the same five detunings. As δ approaches the SP, the peak-to-peak output voltage increases and reaches a maximum of 6.4×10^{-4} V at the SP. **c**, Brillouin response amplitude measured by sweeping the piezoelectric base excitation frequency from 1 Hz to 10 Hz. A signal generator was used to scan a triangular wave function from 1 Hz to 10 Hz over a 300 s interval. **d**, Simultaneously measured displacement signal obtained using a reference interferometer. In **c** and **d**, the shaded frequency interval from 1 Hz to 4.4 Hz marks the range in which the Brillouin-based sensor outperforms the interferometric measurement. **e**, System sensitivity derived by dividing the Brillouin response in **c** by the acceleration obtained from **d**. The shaded region below 1 Hz marks the low-frequency measurement range displayed separately in **f**. The PSD of the system (Supplementary Section 5) reveals multiple mechanical resonances between 10^{-3} Hz and 5 Hz, with a mechanical quality factor $Q_m \approx 7.57$. Measurements are performed at room temperature without vacuum isolation, where pronounced 1/f noise dominates near DC. Owing to multiple resonance-enhanced responses, the peak sensitivity reaches 10^8 V g^{-1}. **f**, Sensitivity between 0.0005 Hz and 1 Hz, calculated from discrete Brillouin-output and reference-interferometer measurements and plotted on logarithmic axes. The minimum measurable frequency is 0.0005 Hz, corresponding to a minimum detectable acceleration of 3.06×10^{-16} g. The noise-equivalent acceleration is 1.17 pg Hz$^{-1/2}$, corresponding to an equivalent displacement noise of 7.85 pm Hz$^{-1/2}$.

separation. The structural PSD (Fig. 4c) shows a mechanical resonance near 0.2 Hz, selected for time-modulated gravitational sensing with a controlled 1.4 µm source displacement.

Finally, to optimize performance by leveraging the core photonics mechanism, we used a BSIA system with a Brillouin frequency of 311.4 MHz and compared the Brillouin response signal under different detunings. As shown in Fig. 4d, the peak-to-peak output signal is maximized at the SP—indicating that the SP's steep phase dispersion is the primary driver of sensing sensitivity. Operating at this SP with dispersion enhancement, we systematically tested a range of lead bead masses (Fig. 4e). We found that our system could still produce a measurable response with a lead bead mass as low as 1.2 g (radius ≈ 3 mm). By leveraging the SP's steep phase dispersion, our system achieves a minimum detectable gravitational-force change of 3.02×10^{-16} N (1.4 µm source displacement). This result, combined with the acceleration sensing data (Fig. 3), demonstrates the use of the SP for weak-force, acceleration and perturbation sensing, including gravitational sensing. To exclude non-gravitational artefacts, we performed control experiments using equal-volume (diameter 5 cm) spheres of different materials (lead, iron, aluminium and polyoxymethylene plastic) under identical conditions (centre distance 4.2 cm, driving displacement 1.4 µm and frequency 0.197 Hz). The measured displacements exhibit a strong linear relationship with the source mass ($R^2 = 0.964$, Pearson's $r = 0.982$), with a fitted slope of 0.01552 ± 0.00212 nm g^{-1} (Supplementary Section 12). This linear mass dependence is expected from Newton's law of gravitation ($\Delta F \propto m_s$) combined with the linear mechanical response of the torsion pendulum ($x \propto \Delta F$), and strongly supports a gravitational origin of the mass signal and is inconsistent with artefacts that depend primarily on volume, surface area, acoustic coupling, air drag or electromagnetic pickup.

Conclusion

The BSIA-induced stopped-light response extends coherent photon–phonon interference beyond its conventional role in slow-light generation and optical delay, establishing it as a practical resource for precision sensing. Whereas previous optomechanical interference studies primarily focused on realizing large optical delays or buffering optical information[36–38], the present work exploits the accompanying steep phase dispersion to transduce minute mechanical perturbations into enhanced optical signals. Within this context,

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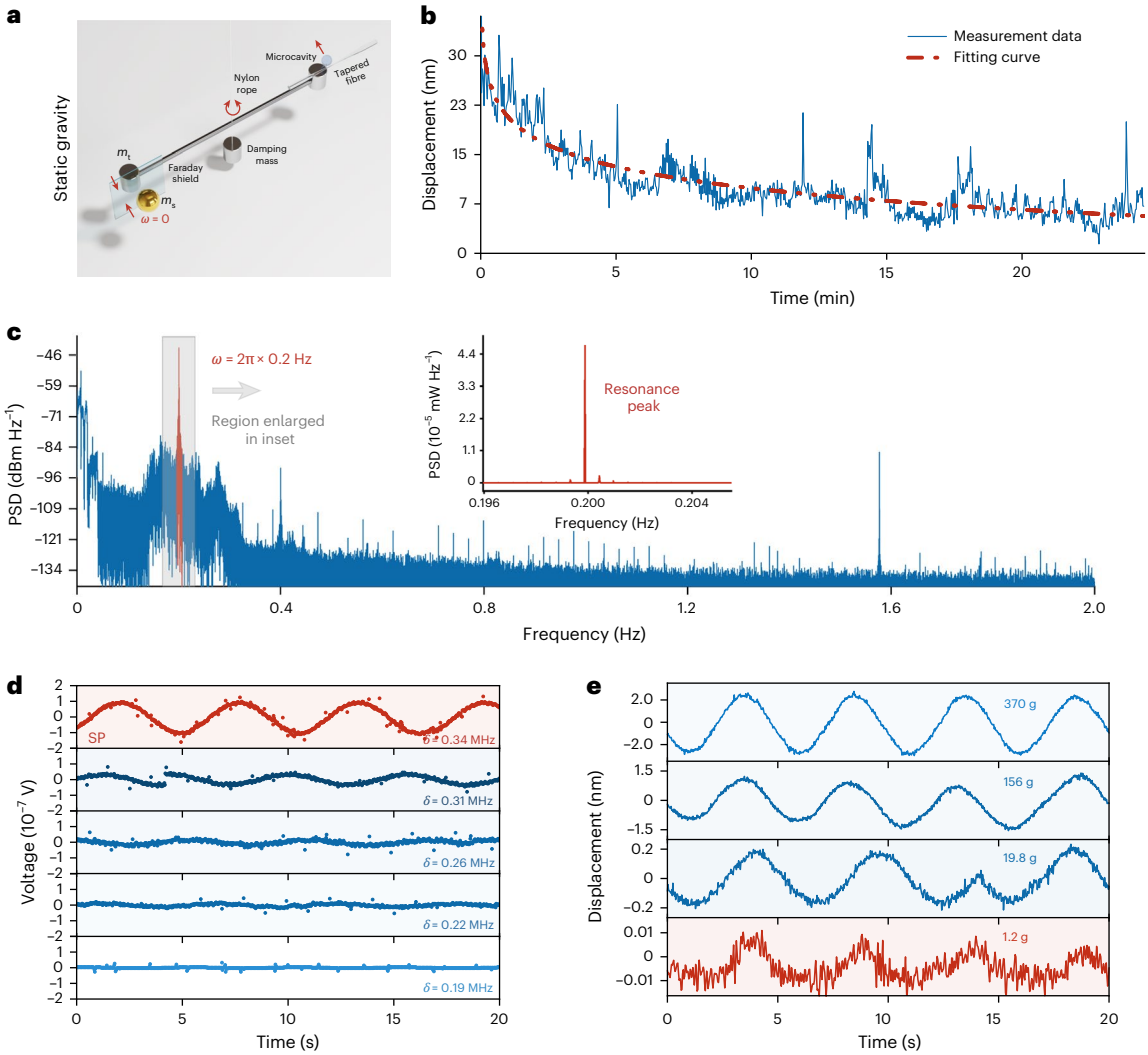


Fig. 4 | Gravity force measurement. a, Experimental set-up for static gravity measurements. The configuration is identical to that used for time-modulated gravitational field measurements (Fig. 1a), except that the lead bead gravitational source is fixed. The test mass experiences a constant gravitational force, causing the optical microcavity to move away from the tapered fibre. The lead bead has a mass of 1.5 kg. **b**, Time evolution of the structural displacement inferred from the Brillouin output signal measured by the VNA. As gravity acts on the torsion pendulum, the microcavity gradually moves away from the tapered fibre, leading to a monotonic signal decrease. The solid trace and dashed curve show the raw data and fitted result, respectively. The initial separation between the test mass and the lead bead is 3 cm, corresponding to an initial static gravitational force of 1.11×10^{-8} N. Over time, the structure undergoes a displacement of 22 nm, corresponding to a gravitational-force change of 1.63×10^{-14} N. **c**, PSD of the torsion-pendulum structure. The pendulum is mounted on a piezoelectric displacement stage. A pronounced mechanical resonance is observed at 0.2 Hz (shaded resonance window). The inset enlarges the shaded resonance window around 0.2 Hz. To maximize sensitivity, a driving frequency of 0.197 Hz is selected. The separation between the test mass (m_t) and source mass (m_s) is 4.2 cm, and the driven displacement of m_s is 1.4 μm. **d**, Verification of SP-enhanced gravity detection. A BSIA feature with a downshifted frequency of 311.4 MHz is selected, and measurements are performed at five different optical detunings (δ). For each detuning, the same source mass (m_s = 40 g) is driven with an identical displacement of 1.4 μm. As the system approaches the SP, the peak-to-peak output signal increases, reaching a maximum at the SP. **e**, Time-resolved structural displacement inferred from the Brillouin signal at the SP (δ = 0.34 MHz) for different source masses. Measurements are performed for m_s ranging from 370 g down to 1.2 g (radius ≈ 3 mm). Below 1.2 g, no discernible response is observed. The static gravitational force between m_t and m_s at 1.2 g and a separation of 4.2 cm is 4.54×10^{-12} N. The minimum detectable gravitational-force change is 3.02×10^{-16} N.

the measured group delay of approximately 420 μs, extracted from the BSIA phase response, is about 8.4×10^3 times larger than the positive delay reported for a representative optomechanical-crystal system[37] and is comparable to the hundreds-of-microseconds regime achieved in room-temperature membrane-in-the-middle platforms[38]. Rather than pursuing long group delay as an end in itself, this work demonstrates its practical value as a sensing resource for displacement, acceleration and gravitational-force transduction.

From the perspective of gravimetry, the present platform complements existing source-mass experiments and inertial sensing approaches by combining high sensitivity with ambient operation. Among the representative acceleration-noise spectral densities[20,27,35] summarized in Supplementary Table 1, the measured sensitivity of $1.17 \times 10^{-12}\ g\ \mathrm{Hz}^{-1/2}$ is more than 2 orders of magnitude below the closest representative value. Importantly, this performance is achieved under room-temperature and atmospheric-pressure conditions, without cryogenic cooling or ultrahigh vacuum, highlighting the practicality of the platform for precision acceleration and gravitational-force sensing.

The present work also differs fundamentally from previous WGM sensing studies in the physical quantity being measured. Earlier WGM demonstrations established highly sensitive optical readout of intrinsic microcavity displacement[41] or measured externally induced

acceleration through mechanical motion[42]. Here the WGM microcavity functions as the optical transducer of a Cavendish-style gravimeter, converting source-mass-induced gravitational perturbations into measurable optical phase shifts through BSIA-enhanced stopped-light dispersion. For the directly comparable acceleration-to-displacement transduction task, the measured displacement-equivalent noise of 7.85 pm $Hz^{-1/2}$ substantially improves upon the representative value of 63.2 pm $Hz^{-1/2}$ reported previously[42], positioning the present platform as a WGM transducer for externally driven acceleration and weak-force detection.

These comparisons show that the present work combines improved sensing performance with a distinct gravimetric architecture that integrates cavity optomechanics with precision gravitational measurements. This Cavendish-style source-mass gravimeter combines a WGM microcavity transducer, BSIA-induced stopped-light dispersion and evanescent nanogap coupling within a single platform. The hierarchical torsion-pendulum architecture bridges macroscopic gravitational interactions and nanoscale microcavity–taper coupling perturbations, while the stopped-light response amplifies their optical readout through steep phase dispersion. With improved environmental isolation and active stabilization of the coupling gap, measurements across systematically varied source masses and separations could test Newtonian scaling and constrain non-Newtonian contributions. Quantitative benchmarking against representative previous studies is provided in Supplementary Tables 1–3 to further illustrate the positioning of the present work.

In summary, this work demonstrates a method for gravitational acceleration sensing by exploiting the steep dispersion at the SP arising from BSIA in a high-Q (~10^8) optical microsphere resonator. Operating at the SP enables access to the maximum phase–frequency slope, which amplifies the optomechanical response to acceleration. The BSIA-induced group delay of approximately 420 μs lowers the light's group velocity to about 2,000 m s^{-1} and thus enhances the optical dispersion. In addition, the exponential dependence of the evanescent field on the coupling distance is leveraged to further enhance the sensing sensitivity. Experiments show that the pronounced dispersion near the SP increases the system's acceleration response. The device achieves a maximum sensitivity of 10^8 V g^{-1} and a minimum detectable acceleration of 3.06×10^{-16} g. The noise-equivalent displacement reaches 7.85 pm/$\sqrt{\text{Hz}}$.

The present implementation is limited by ambient vibration and low-frequency $1/f$ noise under room-temperature, atmospheric-pressure operation, by drift of the ~100 nm microcavity–taper gap and optical detuning, and by a measurement bandwidth set by the torsional resonances. Vacuum operation with improved thermal and seismic isolation, active stabilization of the coupling gap and optical detuning, and engineering of the torsional stiffness and damping should respectively lower the noise floor, improve long-term stability and broaden the operating range. These improvements could extend measurements below the present 1.2 g source-mass limit and to weaker gravitational-force modulations.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41565-026-02298-8.

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Methods

Sample fabrication

The microsphere resonator is fabricated by melting the end of a standard single-mode fibre (G.652D) using an electrode arc discharge. Surface tension reshapes the molten fibre tip into a smooth sphere with a radius of approximately 150 µm. This simple and reproducible process yields optical microcavities with ultrahigh quality factors up to 10^8. After fabrication, an ~1 cm fibre stem is retained to mechanically support the microsphere. The lightweight fibre stem enables attachment of the resonator to the sidewall of an iron column while providing elastic support; upon removal of external acceleration, it supplies a restoring force that returns the microsphere to its equilibrium position.

The tapered fibre used for evanescent coupling is fabricated from standard single-mode fibre by hydrogen-flame heating and simultaneous pulling with motorized translation stages, producing a waist diameter of approximately 1 µm. Both the microsphere and the tapered fibre are mounted on three-dimensional nanopositioning stages (Thorlabs), allowing precise control of spatial alignment and coupling distance.

Observation of the SP

A narrow-linewidth tunable laser (Toptica Photonics) operating near 1,550 nm serves as the control beam to excite forward Brillouin acoustic waves within the microsphere resonator. By tuning the control laser wavelength, the frequency detuning of the Stokes optical mode can be precisely adjusted. An intensity electro-optic modulator (Conquer), driven by a high-speed VNA (Siglent SNA6032A), generates a sideband probe beam before cavity injection. The probe frequency is scanned to resolve BSIA features in the Stokes mode.

The VNA simultaneously records the amplitude, phase and group delay of the transmitted signal. Both control and probe beams are evanescently coupled into the microsphere via the tapered fibre. The output signal is monitored concurrently using a spectrum analyser (Agilent E4440A), the VNA and an oscilloscope, enabling comprehensive characterization of the BSIA spectrum and resonant dispersion properties.

Acceleration measurement

Once the appropriate optical detuning is identified, both the control and probe laser frequencies are fixed. The two transmitted laser fields interfere, producing a beat signal that is detected and fed back into the VNA. The VNA internally mixes this signal with the radio-frequency local oscillator used to modulate the probe beam. A built-in phase-locked loop locks the beat component at the modulation frequency. Following in-phase and quadrature demodulation and low-pass filtering, changes in the optical phase are converted into intensity variations, from which the applied acceleration is extracted.

For signal-to-noise comparisons, driven and undriven traces were acquired at five optical detunings using the same 0.625 Hz, $0.6 \times 10^{-12}\,g$ excitation and identical detection bandwidth. The signal was defined as the peak-to-peak driven response and the noise as the root-mean-square undriven response; their ratio was expressed as $20\log_{10}(V_{pp}/V_{rms})$.

Gravity measurement using a torsion pendulum

Gravitational measurements are performed using a torsion-pendulum configuration analogous to the Cavendish experiment. In contrast to conventional implementations, sensitivity is significantly enhanced by exploiting the exponential dependence of the evanescent coupling rate on the coupling distance, together with the steep phase dispersion near the SP. Coherent detection renders the system highly sensitive to phase variations, which play a central role in gravity sensing.

Static measurements used a fixed 1.5 kg lead source initially 3 cm from the test mass. For time-modulated measurements, the source was mounted on a piezoelectric stage at a 4.2 cm centre-to-centre separation and driven through 1.4 µm at 0.197 Hz. The VNA was operated in single-point mode with a 1 Hz intermediate-frequency bandwidth.

Displacement calibration

Calibration was performed under the laser powers, stopped-light operating point and VNA settings used for the gravity measurements. A 0–50 mV triangular voltage with a 200 s period was applied through a ×10 amplifier to a piezoelectric stage independently calibrated by the interferometer at 1,000 nm V^{-1}. The beat-note amplitude was recorded in single-point mode with a 1 Hz intermediate-frequency bandwidth over 5 cycles. Forward traces were fitted to the full exponential microcavity–taper response, giving a decay constant $\alpha = 0.00833\ \mathrm{nm}^{-1}$ and a characteristic length $1/\alpha \approx 120$ nm. All reported displacements were inferred using this exponential calibration, which agreed with independent interferometric measurements within 5% over the tested range.

Control measurements

Equal-volume, 5-cm-diameter spheres of lead, iron, aluminium and polyoxymethylene were measured at the same 4.2 cm centre distance, 1.4 µm source displacement and 0.197 Hz drive frequency; the source displacement was independently verified for each sphere. Electrostatic shielding was tested with the source mass stationary and a grounded 100 Hz, 5 V_{pp} electric field source placed approximately 1 cm from the aluminium-foil shield; VNA time traces were Fourier transformed to compare the 100 Hz response with and without the shield. Measurements were conducted with the acrylic enclosure closed and the optical-table air suspension activated. Vibration isolation was evaluated interferometrically by applying the same impulsive floor disturbance with the air suspension on and off.

All measurements are conducted under ambient conditions, without vacuum or cryogenic cooling, which ultimately limits the noise floor. Electrostatic shielding is used to suppress spurious electrostatic forces, while a damping mass mounted below the torsion rod improves mechanical stability. The entire apparatus is enclosed within an acrylic shield to minimize air-flow disturbances and mounted on an optical air-floating table to suppress seismic noise.

Data availability

The authors declare that all the data supporting the findings of this study are available within the article and Supplementary Materials files, and also are available from the authors. Source data are provided with this paper.

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Author contributions

W.W. initiated the idea and designed the study. M.T. performed the theoretical study. W.W., X.C. and X.J. supervised the work. Y.Z. performed the experimental work. Q.G., B.X., Y.S., R.Z. and F.A. helped analyse the data and discussion. W.W., Y.Z. and Q.G. wrote the paper. All authors reviewed the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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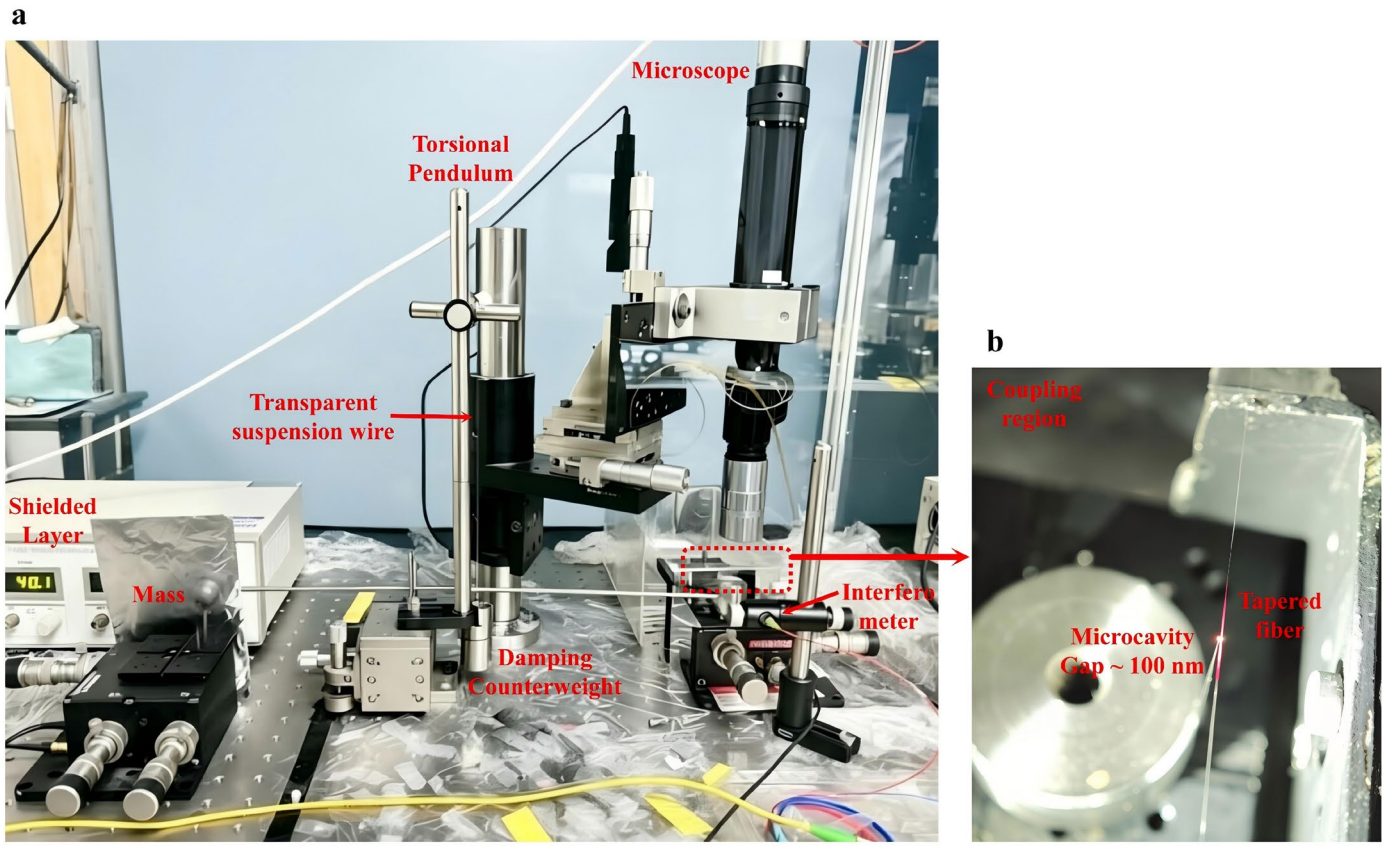
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Extended Data Fig. 1 | Photographs of the experimental setup. a, Photograph of the torsion-pendulum apparatus and optical readout. **b**, Enlarged photograph of the microsphere–taper coupling region, showing the ~100-nm gap.